



Cambridge (CIE) IGCSE Chemistry



Your notes

Preparation of Salts

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Your notes

Preparing Soluble Salts

Preparing soluble salts

What is a salt?

- A salt is a compound that is formed when the **hydrogen atom** in an acid is replaced by a **metal**
- For example if we replace the H in HCl with a potassium atom, then the salt potassium chloride is formed, KCl
- Salts are an important branch of chemistry due to the varied and important uses of this class of compounds
- These uses include fertilisers, batteries, cleaning products, healthcare products and fungicides
- The method used depends on the **solubility** of the salt being prepared

How to name a salt

- The name of salt has two parts
- The first part comes from the **metal**, **metal** oxide or **metal** carbonate used in the reaction
- The second part comes from the **acid**
- The name of the salt can be determined by looking at the reactants
- For example hydrochloric acid always produces salts that end in chloride and contain the **chloride** ion, Cl^-
- Other examples:
 - **Sodium** hydroxide reacts with hydro**chloric** acid to produce sodium chloride
 - **Zinc** oxide reacts with **sulfuric** acid to produce **zinc sulfate**

Preparing soluble salts

- There are two methods of preparing a solution salt:
 - Method **A**
 - Adding acid to a solid metal, insoluble base or insoluble carbonate
 - Method **B**
 - Reacting a dilute acid and alkali (soluble base)

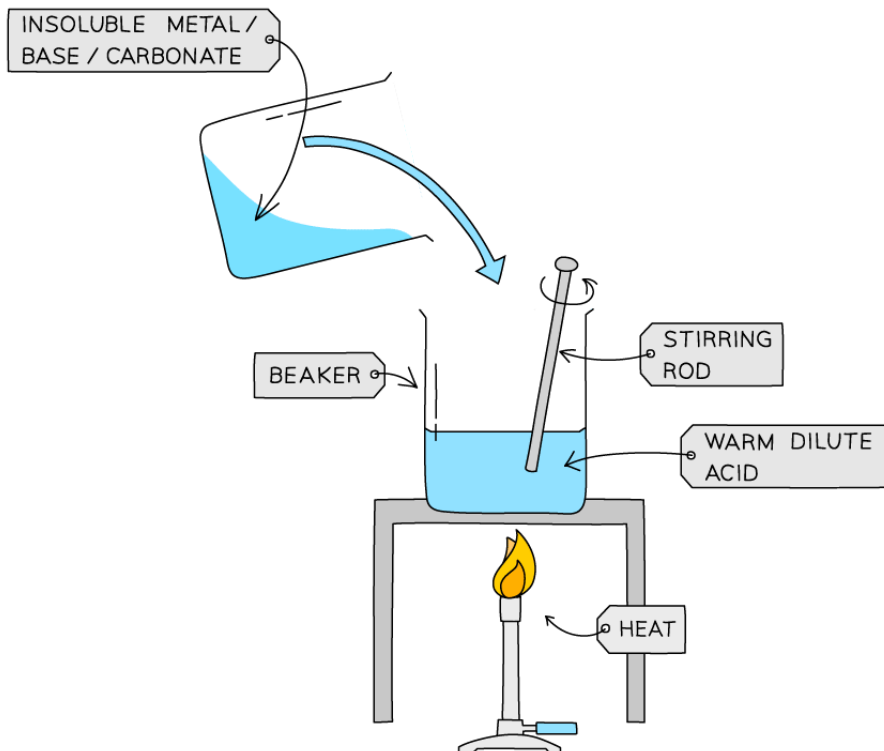
Method A

Method A: Adding acid to a solid metal, insoluble base or insoluble carbonate

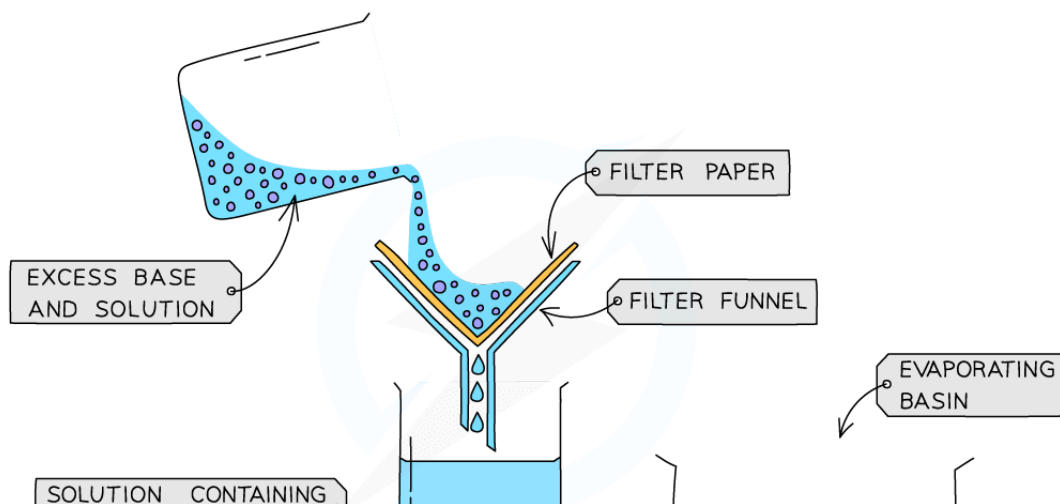


Your notes

- 1 HEAT ACID UNTIL WARM, THEN ADD METAL / BASE / CARBONATE, STIRRING CONSTANTLY UNTIL IT STOPS DISAPPEARING



- 2 FILTER MIXTURE TO REMOVE EXCESS BASE, TRANSFER SOLUTION TO EVAPORATING BASIN.



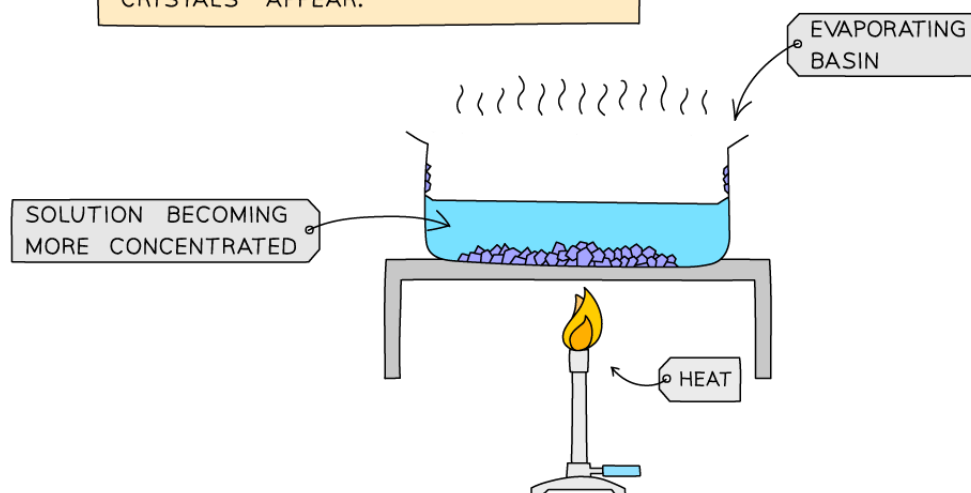


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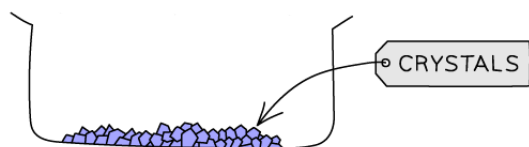
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EVAPORATING WATER FROM SOLUTION USING A BUNSEN BURNER UNTIL CRYSTALS APPEAR.



4

REMOVE EVAPORATING BASIN FROM HEAT AND ALLOW FILTRATE TO DRY AND CRYSTALLISE



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Diagram showing the preparation of soluble salts

Method

- Add dilute acid into a beaker and heat using a Bunsen burner flame
- Add the insoluble metal, base or carbonate, a little at a time, to the warm dilute acid and stir until the base is in excess (i.e. until the base stops disappearing and a suspension of the base forms in the acid)



Your notes

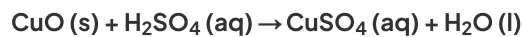
- Filter the mixture into an evaporating basin to remove the excess solid
- Heat the solution to evaporate water and to make the solution saturated. Check the solution is saturated by dipping a cold, glass rod into the solution and seeing if crystals form on the end
- Leave the filtrate in a warm place to dry and crystallize
- Decant excess solution and allow crystals to dry or blot to dry with filter paper

Example: preparation of pure, hydrated copper(II) sulfate crystals using method A

- Add dilute sulfuric acid into a beaker and heat using a Bunsen burner flame
- Add copper(II) oxide (insoluble base), a little at a time to the warm dilute sulfuric acid and stir until the copper (II) oxide is in excess (stops disappearing)
- Filter the mixture into an evaporating basin to remove the excess copper(II) oxide
- Leave the filtrate in a warm place to dry and crystallize

- Decant excess solution
- Blot crystals dry with filter paper

copper(II) oxide + sulfuric acid → copper(II) sulphate + water



Method B

Method B: Reacting a dilute acid and alkali (soluble base)

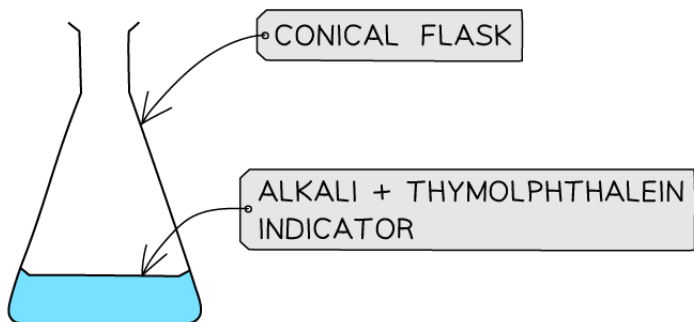


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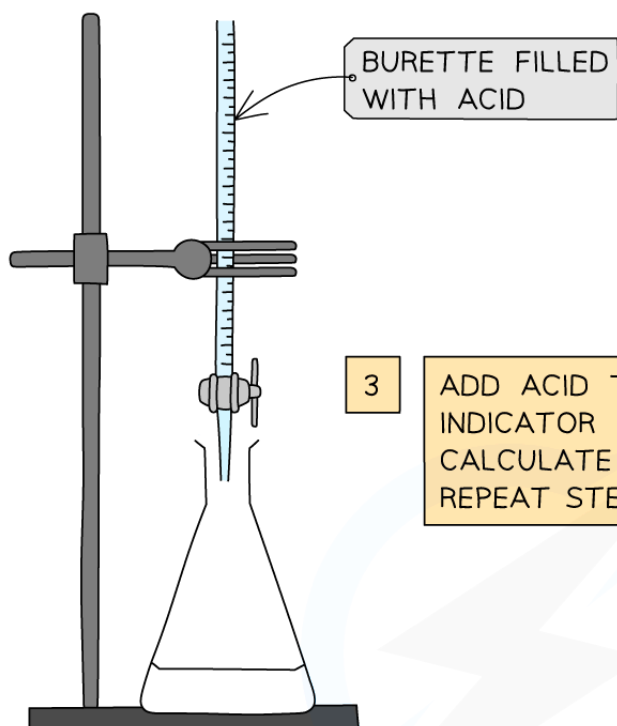


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1 ADD ALKALI + INDICATOR TO CONICAL FLASK USING A PIPETTE



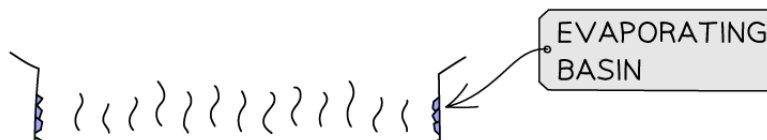
2 ADD ACID TO BURETTE, NOTING THE STARTING VOLUME

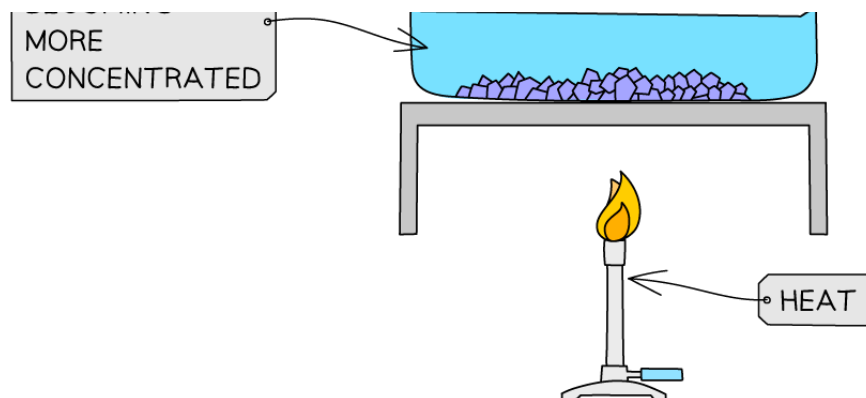


3 ADD ACID TO ALKALI SLOWLY UNTIL INDICATOR CHANGES COLOUR. CALCULATE VOLUME OF ACID ADDED. REPEAT STEPS 1-3 WITHOUT INDICATOR

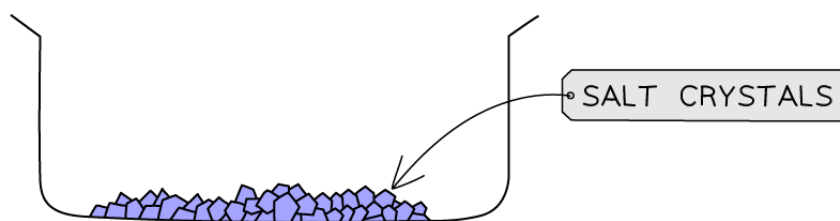
4 TRANSFER SOLUTION TO AN EVAPORATING BASIN, HEAT TO PARTIALLY EVAPORATE WATER

SOLUTION BECOMING





5 REMOVE EVAPORATING BASIN FROM HEAT AND ALLOW FILTRATE TO DRY AND CRYSTALLISE



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Diagram showing the apparatus needed to prepare a salt by titration

Method

- Use a pipette to measure the alkali into a conical flask and add a few drops of indicator (thymolphthalein or methyl orange)
- Add the acid into the burette
- Record the starting volume of acid in the burette
- Add the acid very slowly from the burette to the conical flask until the indicator changes to the appropriate colour
- Record the final volume of acid in the burette
- Calculate the volume of acid added
 - Final volume of acid - initial volume of acid
- Add this same volume of acid to the same volume of alkali without the indicator
- Heat the resulting solution in an evaporating basin to partially evaporate, leaving a saturated solution (crystals just forming on the sides of the basin or on a glass rod dipped in and then removed)

- Leave to crystallise, decant excess solution and allow crystals to dry



Your notes

Preparing Insoluble Salts



Your notes

Preparing insoluble salts

Extended tier only

- Insoluble salts can be prepared using a **precipitation reaction**
- The solid salt obtained is the precipitate, thus in order to successfully use this method the solid salt being formed must be insoluble in water, and the reactants must be soluble

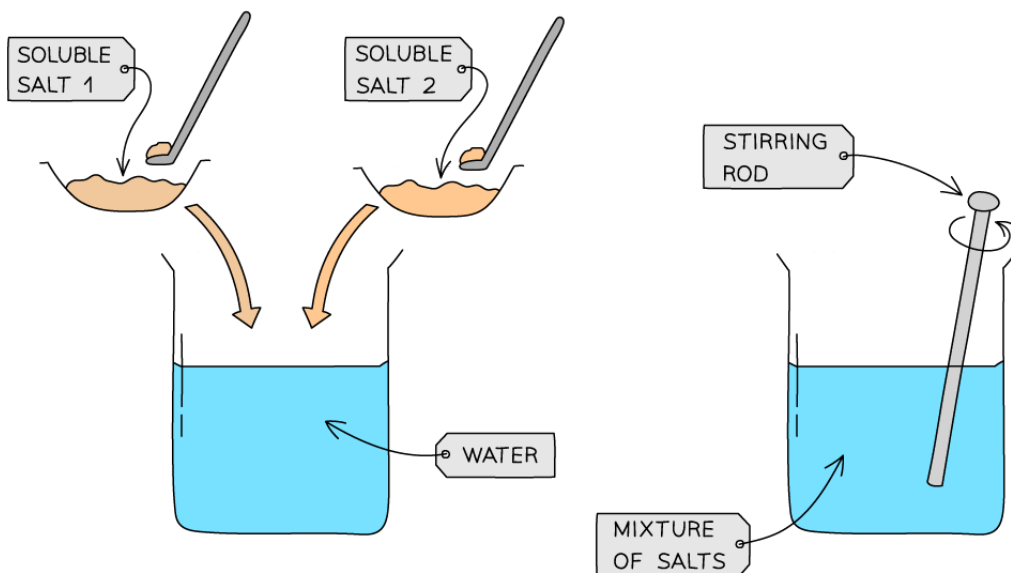
Using two soluble reactants



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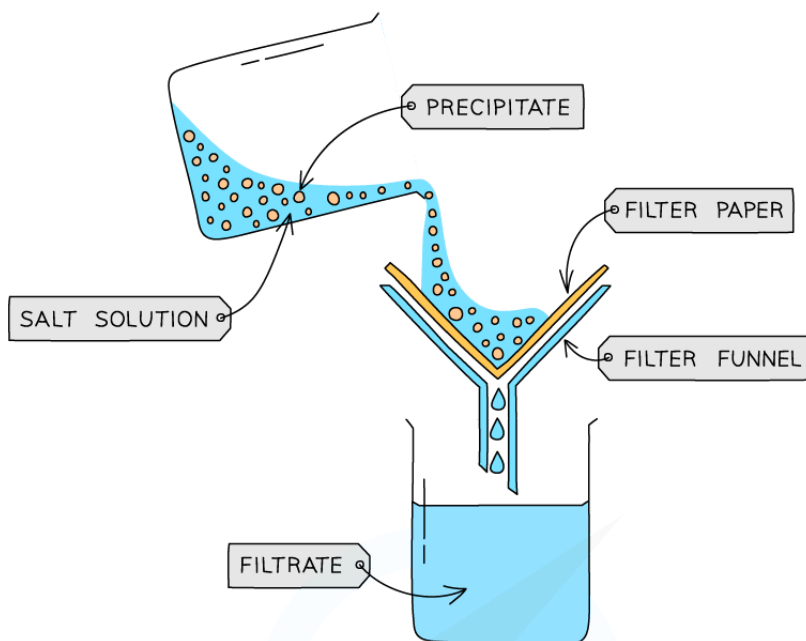
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ADD SOLUBLE SALTS TO WATER AND MIX



2

FILTER TO REMOVE PRECIPITATE FROM THE MIXTURE

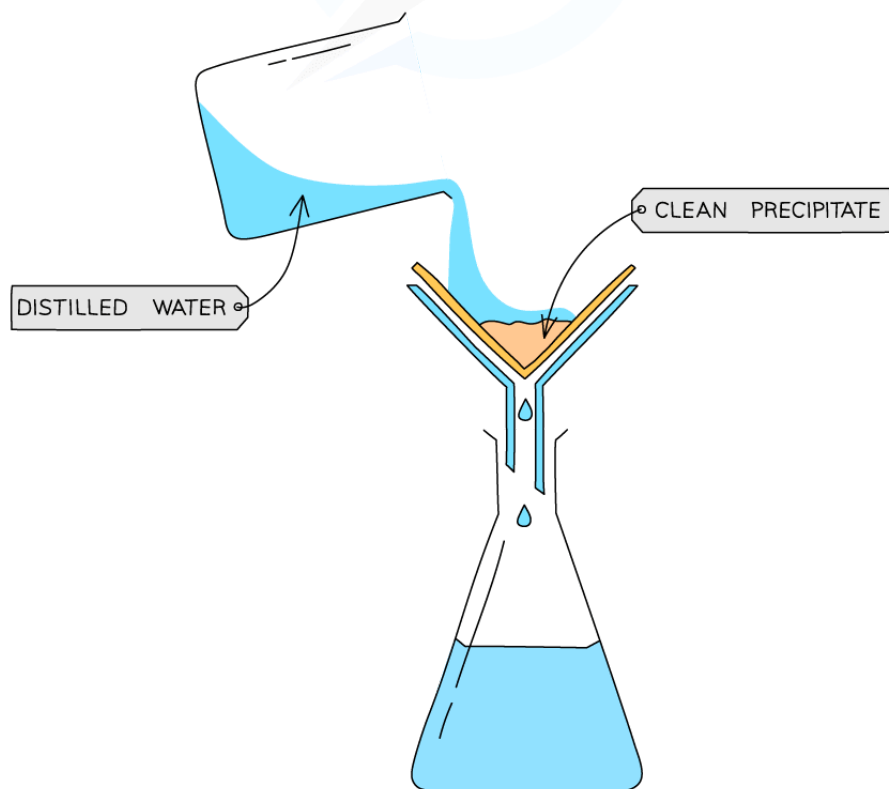




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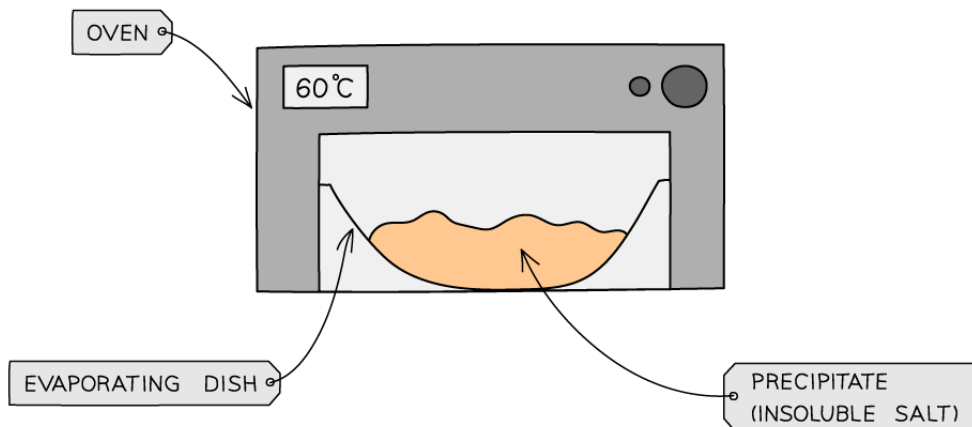
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WASH THE PRECIPITATE WITH DISTILLED WATER TO REMOVE TRACES OF SOLUTION



4

DRY THE PRECIPITATE (INSOLUBLE SALT) IN AN OVEN





Your notes

Diagram showing the filtration of the mixture to remove the precipitate

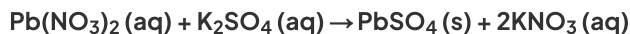
Method

- Dissolve soluble salts in water and mix together using a stirring rod in a beaker
- Filter to remove precipitate from mixture
- Wash the residue with distilled water to remove traces of other solutions
- Leave in an oven to dry

Example: Preparation of pure, dry lead(II) sulfate crystals using a precipitation reaction

- Dissolve lead(II) nitrate and potassium sulfate in water and mix together using a stirring rod in a beaker
- Filter to remove precipitate from mixture
- Wash precipitate with distilled water to remove traces of potassium nitrate solution
- Leave in an oven to dry

lead(II) nitrate + potassium sulfate → lead(II) sulfate + potassium nitrate





Your notes

Solubility Rules

Solubility rules

- Salts are prepared by different methods, depending on whether the salt is soluble or insoluble so it is important to know the solubility of salts

Solubility of the common salts

Salts	Soluble	Insoluble
Sodium, potassium and ammonium	All	None
Nitrates	All	None
Chlorides	Most are soluble	Silver and lead(II)
Sulfates	Most are soluble	Barium, calcium and lead(II)
Carbonates	Carbonates of sodium, potassium and ammonium	Most are insoluble
Hydroxides	Hydroxides of sodium, potassium and ammonium (calcium hydroxide is sparingly soluble)	Most are insoluble



Your notes

Hydrated & Anhydrous Salts

Hydrated & anhydrous salts

- When salts are being prepared, some water can be retained within the structure of the salt during the crystallisation process
- Salts that contain water within their structure are called **hydrated** salts
- Anhydrous** salts are those that contain no water in their structure
- A common example is copper(II) sulfate which crystallises forming the salt hydrated copper(II) sulfate, which is blue
- When it is heated, the water from its structure is removed, forming anhydrous copper(II) sulfate, which is white
- The hydrated salt has been **dehydrated** to form the anhydrous salt
- This reaction can be reversed by adding water to anhydrous copper(II) sulfate



Water of crystallisation

Extended tier only

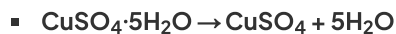
- Water molecules included in the structure of some salts during the crystallisation process are known as **water of crystallisation**
- A compound that contains water of crystallisation is called a **hydrated compound**
- When writing the chemical formula of hydrated compounds, the water of crystallisation is separated from the main formula by a **dot**:
 - Hydrated copper(II) sulfate is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 - Hydrated cobalt(II) chloride is $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$
- The formula shows the number of moles of water contained within one mole of the hydrated salt:
 - Hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, contains 5 moles of water in 1 mole of hydrated salt
- A compound which doesn't contain water of crystallisation is called an **anhydrous compound**:
 - Anhydrous copper(II) sulfate is CuSO_4
 - Anhydrous cobalt(II) chloride is CoCl_2

- The conversion of anhydrous compounds to hydrated compounds is **reversible** by heating the hydrated salt:

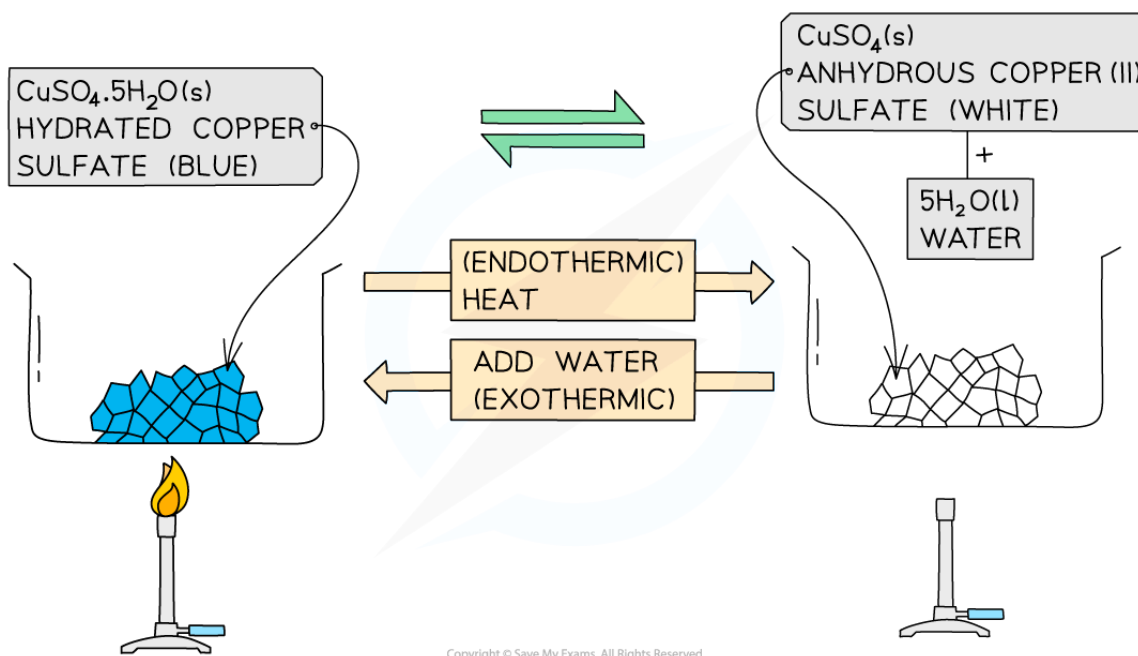
- Anhydrous to hydrated salt:



- Hydrated to anhydrous salt (by heating):



Hydrated copper(II) sulfate and anhydrous copper(II) sulfate



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Diagram showing the dehydration of hydrated copper(II) sulfate



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